



Arab Academy for Science and Technology and Maritime Transport
College of Computing and Information Technology

Course	Introduction to Artificial Intelligence - CAI3101
Lecturer	Dr. Mohamed Farouk

Programming Assignment (2)
Deadline Week13

Part 1 (Implement by your preferred programming language)

- 1) Implement a code for testing a **single perceptron** for a given dataset. The code should accept as input from the user the weights and bias for linear classification. The code should also plot the dataset and the decision boundary.
- 2) By using the function in part(1), implement a code for testing a **Multi-layer of Perceptrons** considering the XOR problem as a non-linear classification problem. The code should accept the weights and bias for each neuron and printout the testing tables at each layer.
- 3) Implement the **Perceptron Learning Algorithm** for a 2D example data. The weights can be initialized by a random solution. The code should be generic for multi-dimensional data sets.

Part 2 (Implement using python)

- 4) You are provided with a sample of the **KDD Dataset** containing **200 records**. Each record represents a network connection with various features. The dataset has a **class label** that indicates whether the connection is **normal** or **anomaly**. Your tasks are as follows:
 - a. **Dataset Splitting:**
Split the dataset randomly into **70% training** and **30% testing** sets. Printout the number of normal and anomaly patterns in both datasets. The splitting should balance the number of patterns between the two classes.
 - b. Train a **Single Perceptron** on the training data and evaluate its performance on the testing data by measuring the accuracy of the classification process.
 - c. Apply the **K-Means Clustering** on the entire dataset using different number of clusters (e.g. k= 2,3,4,5). For each value of k, calculate and report the **percentage of normal and anomaly records** in each cluster.

	Q1	Q2	Q3	Q4.a	Q4.b	Q4.c	Discussion
Marks	2	2	2	2	2	2	8